INCH-POUND

MIL-O-44191A
27 January 1989
SUPERSEDING
MIL-O-44191
28 June 1985

MILITARY SPECIFICATION

OVERALLS, COLD WEATHER, SYNTHETIC FIBER PILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pair of synthetic fiber pile overalls used as a component of the Extended Cold Weather Clothing System.

1.2 Classification. The overalls shall be of one type in the following sizes as specified (See 6.2).

Size	Short/Regular	Long
X-Small	X	X
Small	X	X
Medium	X	X
Large	X	X
X-Large	X	X

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
V-F-106	-	Fastener, Slide, Interlocking
V-T-285	-	Thread, Polyester
DDD-L-20	-	Label: For Clothing, Equipage and Tentage (General Use)
PPP-B-26	-	Bag, Plastic (General Purpose)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-T-2283	-	Tape, Textile, Nylon Woven, White or Dyed
MIL-W-4088	-	Webbing, Textile, Woven Nylon
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment, Equipage; General Specification For
MIL-C-43251	-	Cloth, Pile, Synthetic Fiber Pile

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitching

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members For Containers and Pallets
MIL-STD-1491	-	Glossary of Knitting Imperfections

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COLOR ASSOCIATION OF THE UNITED STATES, INC.

Department of Defense Standard Shades for Sewing Threads

(Application for copies should be addressed to the Color Association of the United States, Inc., East 38th Street, New York, NY 10016.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3..

3.2 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variations from the specification may appear in the sample, in which case the specification shall govern.

3.3 Material. It is encouragd that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The cloth for the overall shall be synthetic fiber pile, Brown 463, class 4 conforming to MIL-C-43251.

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3.3.2 Thread, polyester. The thread for seaming and stitching shall conform to type 1, class 1, subclass A, size B of V-T-285. The thread shall be brown C.A. 66069 and shall show good fastness to laundering.

3.3.3 Tape, hanger. The tape for hanger loop at back bib shall be flat nylon, black, 1/2 inch, conforming to MIL-T-2283.

3.3.4 Webbing, nylon. The webbing used to attaching the suspender loops shall be 2 inch width, black, type VIIIb, class 2 conforming to MIL-W-4088.

3.3.5 Webbing, nylon. The webbing used to secure the ends of the fly slide fastener shall be 1-1/4 inch width, black, type III, class 2 conforming to MIL-W-4088.

3.3.6 Fastener, suspender. The suspender fasteners shall be black, 2 inch, quick release model, ITW Nexus SF, or equivalent (see 6.5).

3.3.7 Slide fastener, side closure. The slide fastener for the side closure shall be black with black thong, individual element molded plastic, size MS, type IV, style 8 conforming to V-F-106, except the fastener shall be equipped with 2 single pull sliders, mouth to mouth arrangement, enabling 2 way opening and the finished length of the thong after knotting shall be 3 $\pm$ 1/4 inches.

3.3.8 Slide fastener, fly. The slide fastener for the fly opening shall be black with black thong, nylon or polyester continuous coil or ladder configuration, size MS, type I, style 19 conforming to V-F-106, except that the finished length of the thong after knotting shall be 3 $\pm$ 1/4 inches.

3.3.9 Labels. All overalls shall have a class 1 identification label, a class 2 size label, and a class 3 instruction label conforming to type VI of DDD-L-20, with the exception of a 1/2 inch margin at the top edge of label. The label color shall be Medium Green, C.A. 70034, 70130, or 70131. The labels shall show fastness to laundering. A 1/2 inch border is to be placed on the top edge of the label. The instruction label shall include the following information:

a. Fixed Laundry: The overalls should be laundered utilizing "Natick Formula IV". NOTE: DO NOT USE BLEACH, STARCH OR DRY CLEAN.

b. Field Laundry: The overalls should be laundered utilizing "Formula II" of TM 10-280. NOTE: DO NOT USE BLEACH, STARCH OR DRY CLEAN.

Drying: The overalls should be finished in a tumble-dryer, loaded not more than 2/3 of rated capacity at lowest fabric cycle (do not exceed 90°F). Remove immediately from dryer. Avoid over-drying. NOTE: DO NOT PRESS.

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c. Machine/Hand Laundering. Use delicate/gentle fabric wash cycle or by hand, using cold water (up to 85°F) and cold water laundry detergent. Rinse in clean cold water. NOTE: DO NOT USE BLEACH OR STARCH.

Drying: Tumble dry at lowest fabric cycle, delicate/gentle (do not exceed 90°F). Remove immediately at end of drying. Avoid over-drying. To drip dry, remove from water and place on rust-proof hanger. NOTE: DO NOT PRESS.

DO NOT REMOVE THIS LABEL

"3.3.9.1 Combination size, identification, and instruction label. The identification label, size label, and instruction label may be combined into one label, conforming to class 14. The three labels shall be printed as one continuous label with the size label first and the identification and instruction labels placed below the size label. The size and identification labels may be combined and the contents placed above the instruction label. A space of 1/2 inch minimum shall be maintained between the labels, with a 1/2 inch border across the entire label. In addition, a solid line 1/16 inch minimum width shall extend across the entire label approximately in the middle of the 1/2 inch blank space. The coating requirement shall conform to the class 3 label."

3.3.10 Suspenders. The suspenders shall be South Carolina Elastic Company spandex finish knit elastic, part no. 15034-K-L, 2 inch wide, black, or equivalent (see 6.5).

3.4 Design. The overalls shall be of the bib type, with adjustable suspenders, side full length slide fastener closure, and slide fastener fly opening.

3.5 Patterns. Standard patterns which provide an allowance of 1/2 inch for setting slide fastener at side closure, and 1/2 inch for all seams will be furnished by the Government. The Government patterns shall not be altered in any way and are to be used only as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns.

3.5.1 Pattern parts. The component parts of the overalls shall be cut from the materials specified below and in accordance with the pattern parts indicated.

List of pattern parts

Material	Nomenclature	Parts
Synthetic fiber pile	Front	2
	Back	2

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3.6 Construction. The construction shall conform in all respects to the requirements specified in table I and herein. Figure I is furnished solely for guidance and information. Should variation from the specification appear in figure I, the specification shall govern.

3.6.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the chain or underside of the stitch shall be on the inside of the overalls.

3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin runouts occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Except for prestitching, thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Type 401, 101, 502, 503, and 504 stitching. Thread tension shall be maintained so that there will be no loose stitching. all repairs shall be in accordance with 3.6.1.1.1a. and b. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. Type 301 stitching may be used to repair 401 stitching.

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3.6.2 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the overalls.

3.7 Manufacturing operations requirements. The overalls shall be manufactured in accordance with all operations specified in table I. The contractor is not required to follow the exact sequence of operations.

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TABLE I.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cut overalls. a. Cut the overalls in strict accordance with patterns furnished, which show size, directional lines, and notches for proper assembly of all parts. Parts shall be cut in accordance with marks on patterns. The pile side of the fabric shall be on the inside of the overalls. b. Cut suspender elastic in the following lengths in inches for sizes X-small to X-large respectively: 29, 30, 31, 32, 33. c. Cut hanger tapes 5 1/2 to 6 inches for all sizes. d. Cut suspender loops 2 3/8 to 2 5/8 inches in length. NOTE: Cut hanger tapes, suspender loops, and zipper thongs with hot wire.						
2.	Replacement of damaged parts. Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages as specified in 4.4.2 are excluded and replaced with non-defective and properly matched material.						
3.	Shade and size marking. Mark, ticket or bundle all parts of the overalls to insure a uniform shade and proper assembly throughout the overalls. Any method of marking may be used except:						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Shade and size marking. (cont'd)						
	a. Corrosive metal fastening devices. b. Sew-on tickets. c. Adhesive type tickets which leave an impression or traces of paper or adhesive on the material after removal of the ticket.						
4.	Assemble overalls.						
	a. Overedge stitch around periphery of all pattern pieces, catching size label 4 inches down from top on right back at center seam edge. The identification and care labels shall be caught on the right back at center seam edge, beginning 4 inches below the top edge of the overalls. When a combination size, identification, and instruction label is used, position and stitch as for identification and care label.	502 or 503	EFd-1	8-10	B	B	
	b. Position side closure slide fasteners with right sides together to left/right, fronts/back respectfully. Stitch in place with slide fastener 1/2 to 5/8 inch from top, extending to the bottom of leg, turning in bottom ends of tape.	301	SSa-1	8-10	B	B	
	c. Turn slide fastener right side out and top stitch 1/4 to 3/16 inch from fold.	301	EFa-1	8-10	B	B	
	d. With right sides together, join fronts and back at center seams from tops to marks on patterns.	301	SSa-1	8-10	B	B	
	e. With right sides together, join leg inseams.	301	SSa-1	8-10	B	B	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Assemble overalls. (cont'd)						
	f. Position wrong side of fly slide fastener tape to the outside (with edges of serging even with teeth of slide fastener) of fly opening and stitch with 1/4 inch spaced double row of stitching 1/16 to 1/8 inch from tape edge.	301	LSp-4	8-10	B	B	
	g. With right sides together stitch elastic suspender straps to back bib 1/16 to 1/4 inch from edge.	301	SSa-1	8-10	B	B	
	h. On the outside, position a 1 1/2 to 1 3/4 inch length of securing tape over each end of fly and turning ends under 1/4 to 3/8 inch, stitch on all four sides with a double row of stitches 1/16 to 1/8 inch from edges with approximately 1/4 inch between rows.	301	LSd-1	8-10	B	B	
	i. Edge stitch around top hem 1/16 to 1/8 inch from folded edge and ends.	301	EFa-2	8-10	B	B	
	j. Turn suspender loops toward top and stitch in place running off ends with a minimum of 3 stitches and stitch over loop twice.	301	LSc-2	8-10	B	B	
	k. Thread suspender strap thru the bar connector. With the connector bar pointing down (right sides facing up), thread the suspender strap from the back through the uppermost opening. Insert the strap through the lower opening, coming back up behind the upper bar. Turn the strap end over 1/2 inch then 1/2 inch again and box stitch or stitch with two rows of stitching 1/4 inch apart tacking ends.	301	EFb-2	8-10	B	B	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Assemble overalls. (cont'd)						
	1. Hem top of overalls by turning in edges 1/2 to 5/8 inch and stitching 1/8 to 1/4 inch from edge, folding back bib ends in, and catching slide fastener, hanger loop, and off set portion of suspender loop tape in stitching.	301	EFa-1	8-10	B	B	
	or						
	As an alternate, stitch through the securing tape 1/16 to 1/8 inch from folded edges with a box X pattern 1/16 to 1/8 inch from edges and 1 to 1-1/4 wide inches wide.	301	EFb-1	6-8	B	B	
	NOTE: Operations L and J may be combined into one operation with stitching spaced 1/4 inch apart.						
5.	Clean overalls.						
	a. Trim thread ends and remove loose threads.						
	b. Remove spots, stains and markings.						
	c. Remove shade tickets.						

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3.8 Finished measurements. The measurements of the finished overalls shall conform to the requirements specified in table II. All measurements shall be taken with the slide fasteners closed and the overalls placed smooth and flat on a table.

TABLE II. Finished measurements (in inches)

Size	1/2 chest 1/	Inseam 2/	Outseam 3/
X-Small S./R.	14 1/2	23 5/8	37-1/2
X-Small Long	14 1/2	27 5/8	42
Small S./R.	16 1/2	23 5/8	38
Small Long	16 1/2	27 5/8	42-1/2
Medium S./R.	18 1/2	23 5/8	38-1/2
Medium Long	18 1/2	27 5/8	43
Large S./R.	20 1/2	23 5/8	39
Large Long	20 1/2	27 5/8	43-1/2
X-Large S./R.	22 1/2	23 5/8	39-1/2
X-Large Long	22 1/2	27 5/8	44
Tolerance	+3/4 - 1/2	± 3/4	± 3/4

1/ Half chest measurements taken 1 inch below front top from folded edge to folded edge with slide fastener closed.

2/ Inseam measurements taken from the center of the crotch to the bottom edge of the leg along the seam.

3/ Outseam measurements taken from the top edge of top hem to the bottom edge of the leg parallel with slide fastener.

3.9 Workmanship. The finished overalls shall conform to the quality of product established by this document. The occurrence of defects shall not exceed applicable acceptable quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the visual defects specified in table III and the dimensional requirements specified in table II.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Certification. Unless otherwise specified, a certificate of compliance will be acceptable as evidence that the suspenders and suspender fasteners conform to the requirements specified in 3.2.6 and 3.2.10.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table III. The lot size shall be expressed in units of overalls. The sample unit shall be one overall. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE III. Visual defects

Examine	Defect	Classification Major	Minor
Material defects	Hole, cut, tear, smash, run or dropped stitch 1/	101	
	Needle chew	102	
	Thin or uneven area	103	
	Mend 1/	104	
	Slub (more than twice the size of the normal diameter of the yarn), float or snag 1/		201
	Foreign matter		202
Color and shade	Color not as specified	105	
	Variation in shade within a part or between two or more parts 2/ 3/		203
	Streaks clearly visible 3/		204
	Barre 1/		205
Cleanness	Spot or stain more than 1/4 inch in largest dimension		206
	Five or more thread ends not trimmed to 1/2 inch or less on outside		207
	Nine or more thread ends not trimmed on inside		208
	One or more shade tickets not removed	209	
	One or more size or shade markings clearly visible on outside 3/		210
Component and assembly	Any component part damaged (unless otherwise specified herein	106	
	Any component part or required operation omitted or added (unless otherwise classified herein)	107	
	Any operation not performed as specified (unless otherwise classified herein)		211
Cutting	Any component part not cut in accordance with directional lines as indicated on patterns or in accordance with document requirements	108	

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TABLE III. Visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams and stitching	Any open seam (except on hems)	109	
	Open seam on hems:		
	- 1/4 to 1 inch		212
	- over 1 inch	110	
	NOTE: A seam shall be considered as being open when one or more stitches joining a seam are broken, or when two or more successive skipped stitches occur.		
	Seam or stitch type not as specified	111	
	Parts of overalls badly pleated, caught or twisted in any type of stitching	112	
	Seam twisted or puckered		213
	Thread breaks (all stitch types) backstitched less than 1/2 inch		214
	Ends of all seams and stitchings produced with 301 stitch type not caught in other seams or stitchings, or backtacked less than 1/2 inch		215
	Ends of a continuous line of stitching overlapping less than 1/2 inch		216
	Raw edges (inclusive of missing or incomplete serging):		
	Any on outside	113	
	On inside 1/8 to 1/2 inch (inclusive)		217
	On inside, over 1/2 inch	114	
	Runoff, or any line of stitching not catching all plies of material to be seamed:		
	- up to 1/2 inch (inclusive)		218
	- more than 1/2 inch	115	
	Stitch tension:		
	Loose tension resulting in a loose seam	116	
	Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	117	
	Up to two stitches per inch less than specified minimum		219
	Three or more stitches per inch less than specified minimum or more than specified maximum	118	

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TABLE III. Visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Stitch tension on overedge	Loose tension in any area		
	- more than 1 inch but not more than 2 inches		220
	- more than 2 inches	119	
Hems	Omitted	120	
Slide fastener	Puckers or pleats around fastener		
	- affecting function	121	
	- not affecting function		221
	Fastener top stop more than 1/4 inch from top		222
	Fastener not working properly	122	
	Damaged	123	
	Thongs omitted	124	
Label	Missing, incorrect, illegible, misplaced, or not attached as specified	125	

1/ For knitted fabric imperfection defects, see MIL-STD-1491.

2/ All parts shall be examined with wales and courses running in same direction.

3/ At normal inspection distance (approximately 3 feet).

4.4.3 End item dimensional examination. The end items shall be examined for conformance to the measurements specified in table II. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of overalls. The sample unit shall be one overall. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container. Open and non-continuous heat sealed seams and closure of polyethylene bag, omission of vent hole in polyethylene bag.
Content	Number of unit packs per shipping container is more or less than specified.

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defects
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial (see 6.2).

5.1.1 Level A. All overalls shall be laid flat with the straps tucked inside and all fasteners closed, then folded to the front, leg over leg. The overalls shall then be folded again, top to bottom, before being placed in a

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close-fitting clear polyethylene bag conforming to A-A-50083 or type I or II, style 1 of PPP-B-26. The bag shall have a 1/4 inch hole located near a bottom corner to allow air to escape.

5.1.2 Commercial. Overalls shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Six overalls of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The shipping container shall be fitted with a box liner conforming to class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, one in length, one in width and six in depth. Approximate dimensions of the container are 21 inches in length, 13 inches in width and 14 inches in depth. Dimensions are furnished as a guide only. The container shall be closed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F load.

5.2.2 Level B packing. Six overalls of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, class domestic, grade 275, variety SW of PPP-B-636. The shipping container shall be fitted with a box liner conforming to class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, one in length, one in width and six in depth. Approximate dimensions of the container are 21 inches in length, 13 inches in width, and 14 inches in depth. Dimensions are furnished as a guide only. The container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Overalls of one size only, preserved as specified in 5.1, shall be packed in accordance with ASTM D-3951.

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"5.3 Palletization. When specified (see 6.2), overalls packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type IA of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV, of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 5 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required information legibly printed or stamped in white directly on the bag across the center face or on a white paper label with black type inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

Stock number
Nomenclature
Size
Quantity

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. The overalls are intended for use as a component of the Extended Cold Weather System.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Size (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.5 Source of supply. Sources of suggested supply for the suspenders and the suspender fasteners are:

1TW
230 Gerry Drive
Wood Dale, Illinois 60101

South Carolina Elastic Co.
Old Howard Gape Rd.
P.O. Box 837
Landrum, South Carolina 29356

6.6 Subject term (key word) listing.

Clothing system ECWCS

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - NU

Review activities:
Army - MD
Navy - MC
DLA - CT

Preparing activity:
Army - GL

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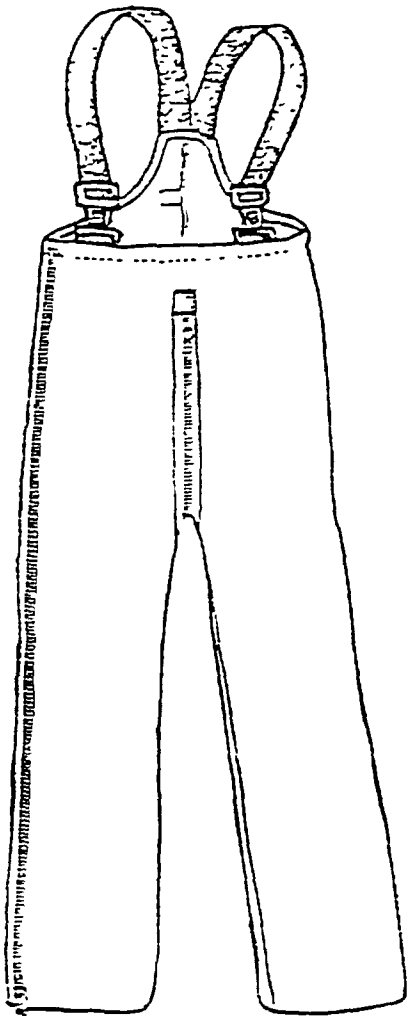


Figure 1: Overalls, Cold Weather, Synthetic Fiber Pile

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-44191A	Overalls, Cold Weather, Synthetic Fiber Pile

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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